Department of Computer Engineering

National Institute of Technology Kurukshetra

MTE – II | April 2025

Subject Name – Big Data Analytics | Subject Code – CSPC-302/ITPC 302

Total Marks – 15 | Duration – 50 Minutes

Q. 1 An online retail company is going to design an application where trending items from different categories can be checked. Let us say a user wants to check what is trending in sports section then trending items from sports only be given. Employ the concept of decaying window in this scenario and propose a possible solution along with its working. Further analyze the effect of number of categories on the performance of solution. Is there any upper limit on the number of news that can be given as input to your system? 06

Q.2 Is fuzzy clustering appropriate for data streams? If yes then justify your answer. Explain the weight update process for fuzzy c-means clustering. Analyze the effect of value of parameter p on behavior of fuzzy c-means clustering. 06

Q.3 What is the role of subspace clustering? Discuses the steps of CLIQUE clustering algorithm briefly. 03

OR

Assume you have six text files, and each file contains a list of products and their corresponding sales figures, represented as two columns: the product name and the sales amount recorded for a given period. It is important to note that each file may contain multiple records for the same product representing different sales events over time. Is it possible to calculate the total sales for each product across all six files using Map-Reduce? If not, why? If yes then show all the steps involved in the process of applying map- reduce. Assume dummy data in the text files.